Example 1

A shopkeeper buys cameras at €180 and sells them at €216 each.
Find his percentage profit on the cost price.

Profit = €216 − €180 = €36

$$\text{Percentage profit} = \frac{\text{Profit}}{\text{Cost price}} \times \frac{100}{1}\%$$

$$= \frac{36}{180} \times \frac{100}{1}\% = \frac{36 \times 10}{18} = 20\%$$

Example 2

By selling a jacket for €138, a store makes a profit of 15% on the cost price.
Find what the store paid for the jacket.

Since a profit of 15% was made, €138 represents 115% of the cost price.

The cost price is 100%.

$$\therefore\ 115\% = €138$$

$$1\% = \tfrac{138}{115} = €1.20$$

$$100\% = 1.20 \times 100 = €120$$

$\therefore$ the store paid €120 for the jacket

Exercise 7.5

1. Find the selling price in each of the following:

(i) Cost price €20: sold at a profit of 20%
(ii) Cost price €32: sold at a profit of 25%
(iii) Cost price €80: sold at a profit of 15%
(iv) Cost price €60: sold at a loss of 10%
(v) Cost price €8.20: sold at a loss of 15%
(vi) Cost price €120: sold at a profit of $12\frac{1}{2}$%.

2. Find the percentage profit or loss in each of the following:

(i) Cost price €8: selling price €10
(ii) Cost price €20: selling price €24
(iii) Cost price €60: selling price €63
(iv) Cost price €120: selling price €96
(v) Cost price €240: selling price €312
(vi) Cost price €16: selling price €12.80.

3. A shopkeeper buys washing machines at €520 each and sells them at €598.
Find his percentage profit.

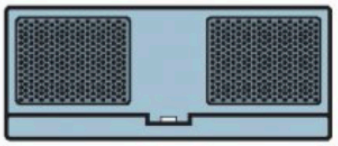
4. Find the four matching pairs out of the following:

5. Find the cost price for each of these if

(i) the selling price is €10 and there is a 25% profit
(ii) the selling price is €8.80 and there is a 10% profit
(iii) the selling price is €57.60 and there is a 20% loss
(iv) the selling price is €22.80 and there is a 5% loss
(v) the selling price is €1050 and there is a 40% profit.

6. A jeweller bought a gold ring for €550 and sold it at a profit of 40%.
Find the sale price.

7. A dealer purchased a used car for €9000 and sold it for €9900.
Find his percentage profit.



8. An electrical shop buys MP3 docking stations at €360 each.
If they are sold at a profit of 15%, find the selling price.

9. A greengrocer buys kiwis in boxes of 50 at €24 per box.
If she sells the kiwis at 60 cent each, find her percentage profit.

10. A fashion shop makes a profit of 30% by selling a coat for €195.
Copy and complete the following to find the cost price of the coat

$$130\% = €195$$

$$1\% = €\frac{195}{130} = \boxed{€\quad\quad}$$

$$100\% = \boxed{€\quad\quad} \times 100 = \boxed{€\quad\quad}$$

11. When an item is sold for €121, a profit of 10% is made.
Find the cost price of the item.

12. By selling a bicycle for €720, a shopkeeper makes a loss of 10%.
(i) What did the shopkeeper pay for the bicycle?
(ii) At what price should he sell the bicycle to make a profit of 15%?

13. A dealer bought television sets at €800 each and sold them at a profit of 40%.
During a sale, the selling price was reduced by 10%.
(i) Find the selling price of a television during the sale.
(ii) What was the dealer's percentage profit on television sets during the sale?

14. A shopkeeper bought 50 toys at €25 each.
He sold 40 of them at a marked price of €32.50.
He sold the remaining 10 toys after reducing the marked price by 20%.
(i) Find what the shopkeeper paid for the 50 toys.
(ii) Find the selling price of each of the 10 remaining toys during the sale.
(iii) Find his total profit on the deal.
(iv) Work out his percentage profit.

15. During a sale, the price of shirts is reduced by 20%.
If the sale price of a shirt is €48, what was the marked price before the sale?

16. A trader bought an article for €250 and then added on 20% to get his selling price.
(i) Find his selling price.
(ii) During a sale, he reduced his selling price by 15%.
Find his percentage profit on the article during the sale.

17. A car dealer bought two cars; a Fiat for €8400 and a Ford for €9600.
He sold the Fiat at a profit of 20% and the Ford at a loss of 5%.
(i) Find the profit he made on the cars.
(ii) Find his percentage profit on the whole deal.